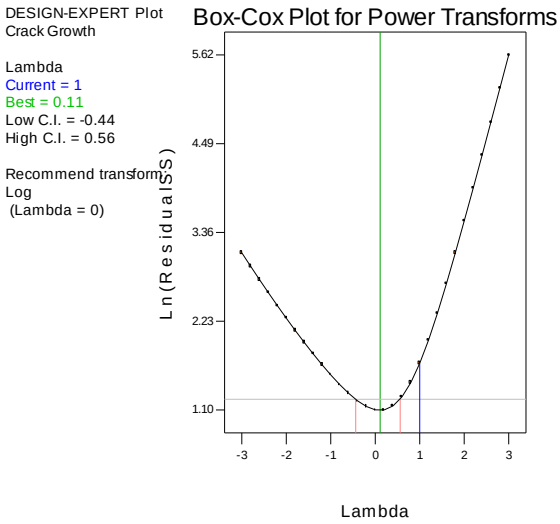
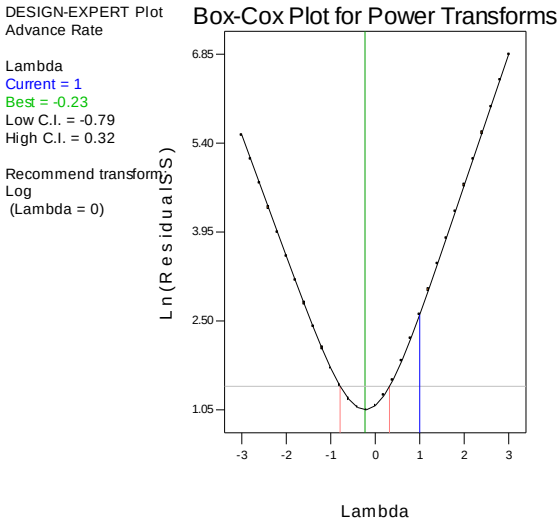
Chapter 15
Other Design and Analysis Topics
Solutions

15.1S. Reconsider the experiment in Problem 5.25. Use the Box-Cox procedure to determine if a transformation on the response is appropriate (or useful) in the analysis of the data from this experiment.



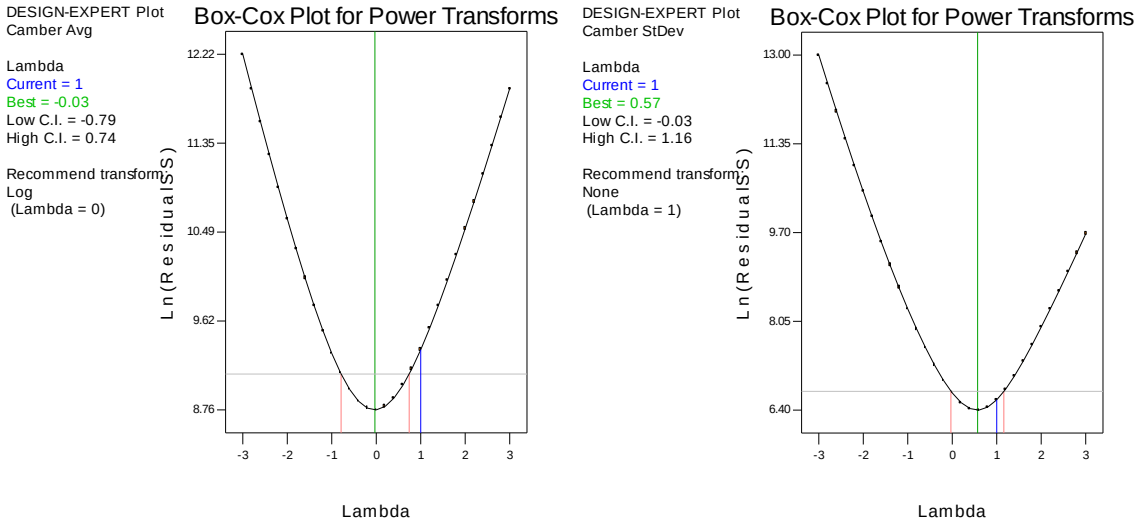
With the value of lambda near zero, and since the confidence interval does not include one, a natural log transformation would be appropriate.

15.2S. In Example 6.3 we selected a log transformation for the drill advance rate response. Use the Box-Cox procedure to demonstrate that this is an appropriate data transformation.



Because the value of lambda is very close to zero, and the confidence interval does not include one, the natural log was the correct transformation chosen for this analysis.

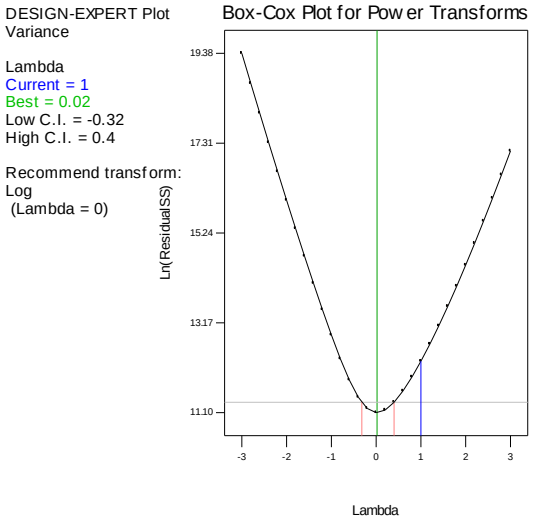
15.3. In Problem 8.22 a replicated fractional factorial design was used to study substrate camber in semiconductor manufacturing. Both the mean and standard deviation of the camber measurements were used as response variables. Is there any indication that a transformation is required for either response?



The Box-Cox plot for the Camber Average suggests a natural log transformation should be applied. This decision is based on the confidence interval for lambda not including one and the point estimate of lambda being very close to zero. With a lambda of approximately 0.5, a square root transformation could be considered for the Camber Standard Deviation; however, the confidence interval indicates that no transformation is needed.

15.4. Reconsider the photoresist experiment in Problem 8.23. Use the variance of the resist thickness at each test combination as the response variable. Is there any indication that a transformation is required?

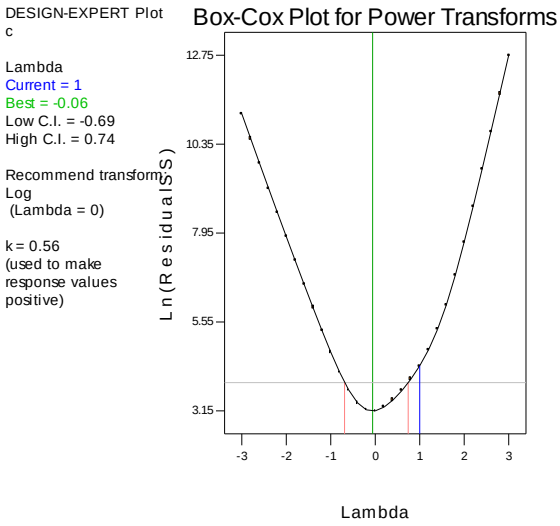
Table P8.4 from Problem 8.23 presents the range, and not the variance. The variance must first be calculated, and the most appropriate model fit to the data. A Box-Cox plot is shown below.



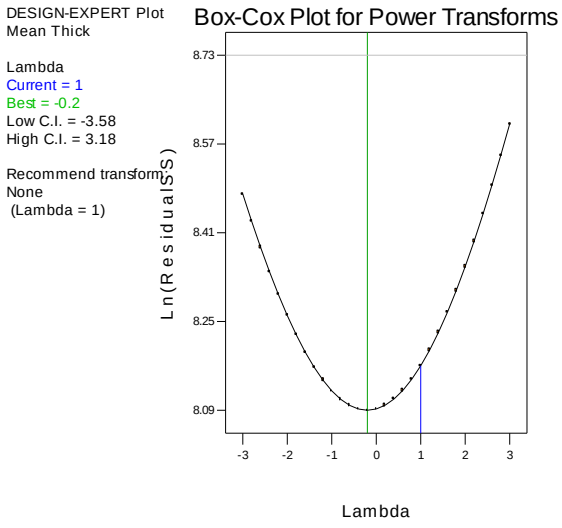
With the point estimate of lambda near zero, and the confidence interval for lambda not inclusive of one, a log transformation would be appropriate. The parameters included in the model affect this plot.

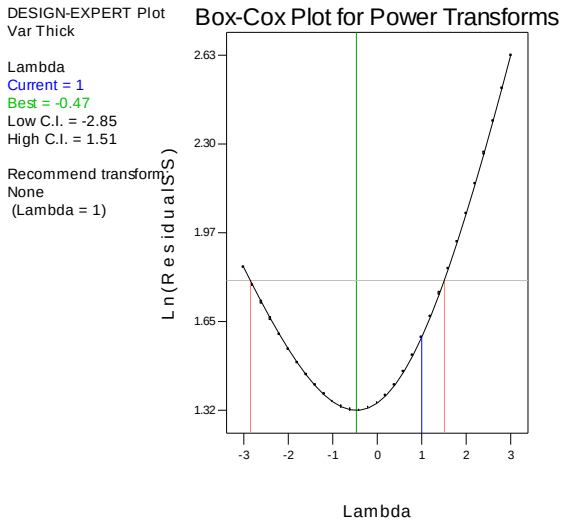
15.5S. In the grill defects experiment described in Problem 8.41 a variation of the square root transformation was employed in the analysis of the data. Use the Box-Cox method to determine if this is the appropriate transformation.

The Box-Cox plot is shown below. Because the confidence interval for the minimum lambda does not include one, the decision to use a transformation is correct. Because the lambda point estimate is close to zero, the natural log transformation would be appropriate. This is a stronger transformation than the square root.



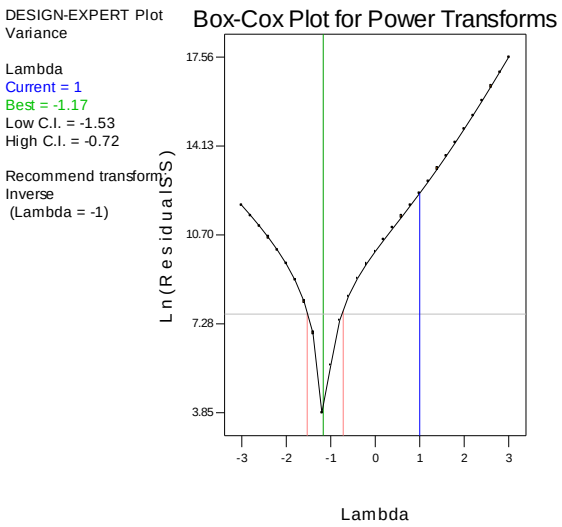
15.6. In the central composite design of Problem 11.11, two responses were obtained, the mean and variance of an oxide thickness. Use the Box-Cox method to investigate the potential usefulness of transformation for both of these responses. Is the log transformation suggested in part (c) of that problem appropriate?





The Box-Cox plot for the Mean Thickness model suggests that a natural log transformation could be applied; however, the confidence interval for lambda includes one. Therefore, a transformation would have a minimal effect. The natural log transformation applied to the Variance of Thickness model appears to be acceptable; however, again the confidence interval for lambda includes one.

15.7. Problem 12.8 suggests using the $ln(s^2)$ as the response (refer to part b). Does the Box-Cox method indicate that a transformation is appropriate?



Because the confidence interval for lambda does not include one, a transformation should be applied. The confidence interval does not include zero; therefore, the natural log transformation is inappropriate. With the point estimate of lambda at –1.17, the reciprocal transformation is appropriate.

15.8. Myers, Montgomery and Vining (2002) describe an experiment to study spermatozoa survival. The design factors are the amount of sodium citrate, the amount of glycerol, and equilibrium time, each at two levels. The response variable is the number of spermatozoa that survive out of fifty that were tested at each set of conditions. The data are in the following table. Analyze the data from this experiment with logistical regression. E – e

Sodium Citrate	Glycerol	Equilibrium Time	Number Survived
–	–	–	34
+	–	–	20
–	+	–	8
+	+	–	21
–	–	+	30
+	–	+	20
–	+	+	10
+	+	+	25

Minitab Output

Binary Logistic Regression: Number Survi, Freq versus Sodium Citra, Glycerol, .

```
Link Function: Logit

Response Information
```

```
Variable        Value    Count
Number Survived  Success    168
                 Failure    232
Freq             Total      400

Logistic Regression Table
                                                  Odds     95% CI
Predictor            Coef   SE Coef      Z      P  Ratio  Lower  Upper
Constant         -0.376962  0.110113  -3.42  0.001
Sodium Citrate    0.0932642  0.110103   0.85  0.397   1.10   0.88   1.36
Glycerol         -0.463247  0.110078  -4.21  0.000   0.63   0.51   0.78
Equilbrium Time   0.0259045  0.109167   0.24  0.812   1.03   0.83   1.27
AB               0.585116  0.110066   5.32  0.000   1.80   1.45   2.23
AC              0.0543714  0.109317   0.50  0.619   1.06   0.85   1.31
BC               0.112190  0.108845   1.03  0.303   1.12   0.90   1.38

Log-Likelihood = -248.028
Test that all slopes are zero: G = 48.178, DF = 6, P-Value = 0.000

Goodness-of-Fit Tests

Method          Chi-Square  DF      P
Pearson           0.113790   1  0.736
Deviance          0.113865   1  0.736
Hosmer-Lemeshow    0.113790   6  1.000
```

This analysis shows that Glycerol (*B*) and the Sodium Citrate x Glycerol (*AB*) interaction have an effect on the survival rate of spermatozoa.

15.9S. The sums of squares and products for a single-factor analysis of covariance follow. Complete the analysis and draw appropriate conclusions. Use $\alpha = 0.05$.

Source of Variation	Degrees of Freedom	Sums of Squares and Products		
		x	*xy*	*y*
Treatment	3	1500	1000	650
Error	12	6000	1200	550
Total	15	7500	2200	1200

		Sums of Squares and Products			Adjusted			
Source	*df*	*x*	*xy*	*y*	*y*	*df*	*MS*	F_0
Treatment	3	1500	1000	650	-	-		
Error	12	6000	1200	550	310	11	28.18	
Total	15	7500	2200	1200	554.67	14		
Adjusted	Treat.				244.67	3	81.56	2.89

Treatments differ only at 10%.

15.10. Find the standard errors of the adjusted treatment means in Example 15.5.

From Example 15.5, $\text{adj}\,\bar{y}_{1.} = 40.38$, $\text{adj}\,\bar{y}_{2.} = 41.42$, $\text{adj}\,\bar{y}_{3.} = 38.80$

$$S_{adj.\bar{y}_{1.}} = \left[2.54 \left\{ \frac{1}{5} + \frac{(25.20 - 24.13)^2}{195.60} \right\} \right]^{\frac{1}{2}} = 0.7231$$

$$S_{adj.\bar{y}_{2.}} = \left[2.54 \left\{ \frac{1}{5} + \frac{(26.00 - 24.13)^2}{195.60} \right\} \right]^{\frac{1}{2}} = 0.7439$$

$$S_{adj.\bar{y}_{3.}} = \left[2.54 \left\{ \frac{1}{5} + \frac{(21.20 - 24.13)^2}{195.60} \right\} \right]^{\frac{1}{2}} = 0.7871$$

15.11. Four different formulations of an industrial glue are being tested. The tensile strength of the glue when it is applied to join parts is also related to the application thickness. Five observations on strength (y) in pounds and thickness (x) in 0.01 inches are obtained for each formulation. The data are shown in the following table. Analyze these data and draw appropriate conclusions.

				Glue Formulation			
1	1	2	2	3	3	4	4
y	*x*	*y*	*x*	*y*	*x*	*y*	*x*
46.5	13	48.7	12	46.3	15	44.7	16
45.9	14	49.0	10	47.1	14	43.0	15
49.8	12	50.1	11	48.9	11	51.0	10
46.1	12	48.5	12	48.2	11	48.1	12
44.3	14	45.2	14	50.3	10	48.6	11

From the analysis performed in *Minitab*, glue formulation does not have a statistically significant effect on strength. As expected, glue thickness does affect strength.

Minitab Output

General Linear Model: Strength versus Glue

```
Factor    Type Levels Values
Glue     fixed     4 1 2 3 4

Analysis of Variance for Strength, using Adjusted SS for Tests

Source    DF    Seq SS    Adj SS    Adj MS      F     P
Thick      1    68.852    59.566    59.566  42.62 0.000
Glue       3     1.771     1.771     0.590   0.42 0.740
Error     15    20.962    20.962     1.397
Total     19    91.585

Term         Coef  SE Coef       T     P
Constant   60.089    1.944   30.91 0.000
Thick     -1.0099   0.1547   -6.53 0.000

Unusual Observations for Strength

Obs  Strength      Fit     SE Fit  Residual   St Resid
  3   49.8000  47.5299     0.5508    2.2701     2.17R

R denotes an observation with a large standardized residual.

Expected Mean Squares, using Adjusted SS

Source      Expected Mean Square for Each Term
 1 Thick    (3) + Q[1]
 2 Glue     (3) + Q[2]
 3 Error    (3)

Error Terms for Tests, using Adjusted SS

Source      Error DF  Error MS  Synthesis of Error MS
 1 Thick       15.00    1.397  (3)
 2 Glue        15.00    1.397  (3)
```

```
Variance Components, using Adjusted SS

Source    Estimated Value
Error             1.397
```

15.12. Compute the adjusted treatment means and their standard errors using the data in Problem 15.11.

$$\text{adj}\,\bar{y}_{i.} = \bar{y}_{i.} - \hat{\beta}\left(\bar{x}_{i.} - \bar{x}_{..}\right)$$

$$\text{adj}\,\bar{y}_{1.} = 46.52 - (-1.0099)(13.00 - 12.45) = 47.08$$

$$\text{adj}\,\bar{y}_{2.} = 48.30 - (-1.0099)(11.80 - 12.45) = 47.64$$

$$\text{adj}\,\bar{y}_{3.} = 48.16 - (-1.0099)(12.20 - 12.45) = 47.91$$

$$\text{adj}\,\bar{y}_{4.} = 47.08 - (-1.0099)(12.80 - 12.45) = 47.43$$

$$S_{adj.\bar{y}_{i.}} = \left[MS_E \left\{ \frac{1}{n} + \frac{\left(\bar{x}_{i.} - \bar{x}_{..}\right)^2}{E_{xx}} \right\} \right]^{\frac{1}{2}}$$

$$S_{adj.\bar{y}_{1.}} = \left[1.40 \left\{ \frac{1}{5} + \frac{(13.00 - 12.45)^2}{58.40} \right\} \right]^{\frac{1}{2}} = 0.5360$$

$$S_{adj.\bar{y}_{2.}} = \left[1.40 \left\{ \frac{1}{5} + \frac{(11.80 - 12.45)^2}{58.40} \right\} \right]^{\frac{1}{2}} = 0.5386$$

$$S_{adj.\bar{y}_{3.}} = \left[1.40 \left\{ \frac{1}{5} + \frac{(12.20 - 12.45)^2}{58.40} \right\} \right]^{\frac{1}{2}} = 0.5306$$

$$S_{adj.\bar{y}_{4.}} = \left[1.40 \left\{ \frac{1}{5} + \frac{(12.80 - 12.45)^2}{58.40} \right\} \right]^{\frac{1}{2}} = 0.5319$$

The adjusted treatment means can also be generated in *Minitab* as follows:

Minitab Output

```
Least Squares Means for Strength

Glue     Mean   SE Mean
1       47.08    0.5355
2       47.64    0.5382
3       47.91    0.5301
4       47.43    0.5314
```

15.13. An engineer is studying the effect of cutting speed on the rate of metal removal in a machining operation. However, the rate of metal removal is also related to the hardness of the test specimen. Five observations are taken at each cutting speed. The amount of metal removed (y) and the hardness of the specimen (x) are shown in the following table. Analyze the data using and analysis of covariance. Use α=0.05.

		Cutting	Speed	(rpm)	
1000	1000	1200	1200	1400	1400
y	x	y	x	y	x
68	120	112	165	118	175
90	140	94	140	82	132
98	150	65	120	73	124
77	125	74	125	92	141
88	136	85	133	80	130

As shown in the analysis performed in *Minitab*, there is no difference in the rate of removal between the three cutting speeds. As expected, the hardness does have an impact on rate of removal.

Minitab Output

General Linear Model: Removal versus Speed

```
Factor    Type Levels Values
Speed    fixed     3 1000 1200 1400

Analysis of Variance for Removal, using Adjusted SS for Tests

Source    DF    Seq SS    Adj SS    Adj MS      F     P
Hardness   1    3075.7    3019.3    3019.3  347.96  0.000
Speed      2       2.4       2.4       1.2    0.14  0.872
Error     11      95.5      95.5       8.7
Total     14    3173.6

Term         Coef   SE Coef       T     P
Constant   -41.656     6.907   -6.03  0.000
Hardness   0.93426   0.05008   18.65  0.000
Speed
1000        0.478     1.085    0.44  0.668
1200        0.036     1.076    0.03  0.974

Unusual Observations for Removal

Obs   Removal      Fit     SE Fit  Residual   St Resid
  8    65.000   70.491      1.558    -5.491    -2.20R

R denotes an observation with a large standardized residual.

Expected Mean Squares, using Adjusted SS

Source      Expected Mean Square for Each Term
 1 Hardness  (3) + Q[1]
 2 Speed     (3) + Q[2]
 3 Error     (3)

Error Terms for Tests, using Adjusted SS

Source      Error DF  Error MS  Synthesis of Error MS
 1 Hardness    11.00      8.7  (3)
 2 Speed       11.00      8.7  (3)

Variance Components, using Adjusted SS

Source   Estimated Value
Error              8.677

Means for Covariates

Covariate     Mean    StDev
Hardness     137.1    15.94

Least Squares Means for Removal

Speed     Mean   SE Mean
1000     86.88     1.325
1200     86.44     1.318
```

```
1400     85.89     1.328
```

15.14S. Show that in a single factor analysis of covariance with a single covariate a 100(1-α) percent confidence interval on the i_{th} adjusted treatment mean is

$$\bar{y}_{i.} - \hat{\beta}\left(\bar{x}_{i.} - \bar{x}_{..}\right) \pm t_{\alpha/2,a(n-1)-1}\left[MS_E\left(\frac{1}{n} + \frac{\left(\bar{x}_{i.} - \bar{x}_{..}\right)^2}{E_{xx}}\right)\right]^{\frac{1}{2}}$$

Using this formula, calculate a 95 percent confidence interval on the adjusted mean of Machine 1 in Example 15.5.

The 100(1–α) percent interval on the i_{th} adjusted treatment mean would be

$$\bar{y}_{i.} - \hat{\beta}\left(\bar{x}_{i.} - \bar{x}_{..}\right) \pm t_{\alpha/2,a(n-1)-1} S_{adj\bar{y}_{i.}}$$

where $\bar{y}_{i.} - \hat{\beta}\left(\bar{x}_{i.} - \bar{x}_{..}\right)$ is an estimator of the i_{th} adjusted treatment mean. The standard error of the adjusted treatment mean is found as follows:

$$V\left(adj.\bar{y}_{i.}\right) = V\left[\bar{y}_{i.} - \hat{\beta}\left(\bar{x}_{i.} - \bar{x}_{..}\right)\right] = V\left(\bar{y}_{i.}\right) + \left(\bar{x}_{i.} - \bar{x}_{..}\right)^2 V\left(\beta\right)$$

Since the $\left\{\bar{y}_{i.}\right\}$ and $\hat{\beta}$ are independent. From regression analysis, we have $V\left(\hat{\beta}\right) = \frac{\sigma^2}{E_{xx}}$.

Therefore,

$$V\left(adj.\bar{y}_{i.}\right) = \frac{\sigma^2}{n} + \frac{\left(\bar{x}_{i.} - \bar{x}_{..}\right)^2 \sigma^2}{E_{xx}} = \sigma^2\left[\frac{1}{n} + \frac{\left(\bar{x}_{i.} - \bar{x}_{..}\right)^2}{E_{xx}}\right]$$

Replacing σ^2 by its estimator MS_E, yields

$$\hat{V}\left(adj.\bar{y}_{i.}\right) = MS_E\left[\frac{1}{n} + \frac{\left(\bar{x}_{i.} - \bar{x}_{..}\right)^2}{E_{xx}}\right] \text{ or}$$

$$S\left(adj.\bar{y}_{i.}\right) = \left\{MS_E\left[\frac{1}{n} + \frac{\left(\bar{x}_{i.} - \bar{x}_{..}\right)^2}{E_{xx}}\right]\right\}^{\frac{1}{2}}$$

Substitution of this result into $\bar{y}_{i.} - \hat{\beta}\left(\bar{x}_{i.} - \bar{x}_{..}\right) \pm t_{\alpha/2,a(n-1)-1} S_{adj\bar{y}_{i.}}$ will produce the desired confidence interval. A 95% confidence interval on the mean of machine 1 would be found as follows:

$$adj.\bar{y}_{i.} = \bar{y}_{i.} - \hat{\beta}\left(\bar{x}_{i.} - \bar{x}_{..}\right) = 40.38$$
$$S\left(adj.\bar{y}_{i.}\right) = 0.7231$$

$$\left[40.38 \pm t_{0.025,11}\left(0.7231\right)\right]$$
$$\left[40.38 \pm \left(2.20\right)\left(0.7231\right)\right]$$
$$\left[40.38 \pm 1.59\right]$$

Therefore, $38.79 \le \mu_1 \le 41.97$, where μ_1 denotes the true adjusted mean of treatment one.

15.15. Discuss how the operating characteristic curves for the analysis of variance can be used in the analysis of covariance.

To use the operating characteristic curves, fixed effects case, we would use as the parameter Φ^2,

$$\Phi^2 = \frac{a\sum \tau_i^2}{n\sigma^2}$$

The test has a–1 degrees of freedom in the numerator and $a(n–1)$ –1 degrees of freedom in the denominator.

15.16. Three different Pinot Noir wines were evaluated by a panel of eight judges. The judges are considered a random panel of all possible judges. The wines are evaluated on a 100-point scale. The wines were presented in random order to each judge, and the following results were obtained.

	Wine		
Judge	1	2	3
1	85	88	93
2	90	89	94
3	88	90	98
4	91	93	96
5	92	92	95
6	89	90	95
7	90	91	97
8	91	89	98

Analyze the data from this experiment. Is there a difference in the wine quality? Analyze the residuals and comment on model adequacy.

This is a repeated measures problem where the three Pinot Noirs are the treatments and the random Judges are the subjects. As described in the textbook, the analysis for a single factor repeated measures is the same as the randomized complete block design, RCBD, where the Judges are the random blocks.

The analysis below identifies a significant difference in the wine quality.

The residual plots do not identify any concerns with model adequacy.

Design Expert Output

Response 1 Wine Quality
ANOVA for selected factorial model
Analysis of variance table [Classical sum of squares - Type II]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F
Block	48.00	7	6.86		

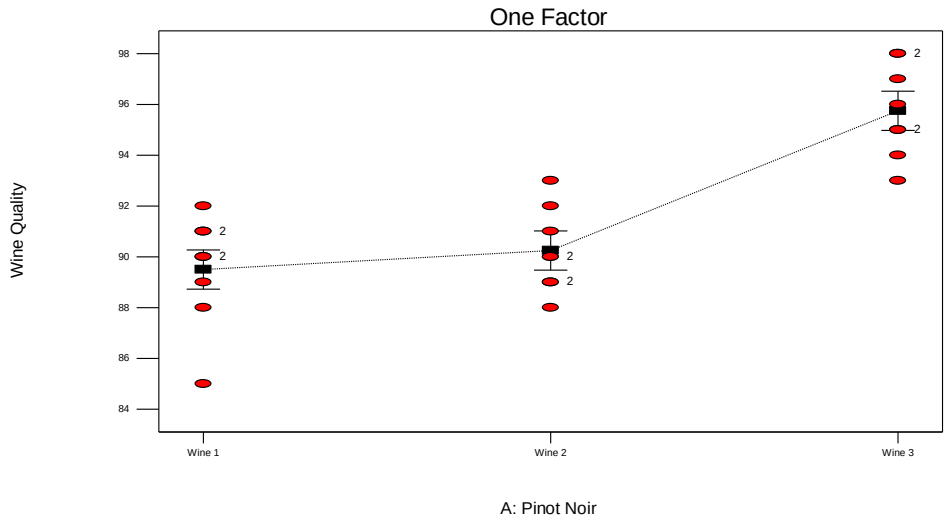
Model	186.33	2	93.17	44.98	< 0.0001	significant
A-Pinot Noir	*186.33*	*2*	*93.17*	*44.98*	*< 0.0001*	
Residual	29.00	14	2.07			
Cor Total	263.33	23				

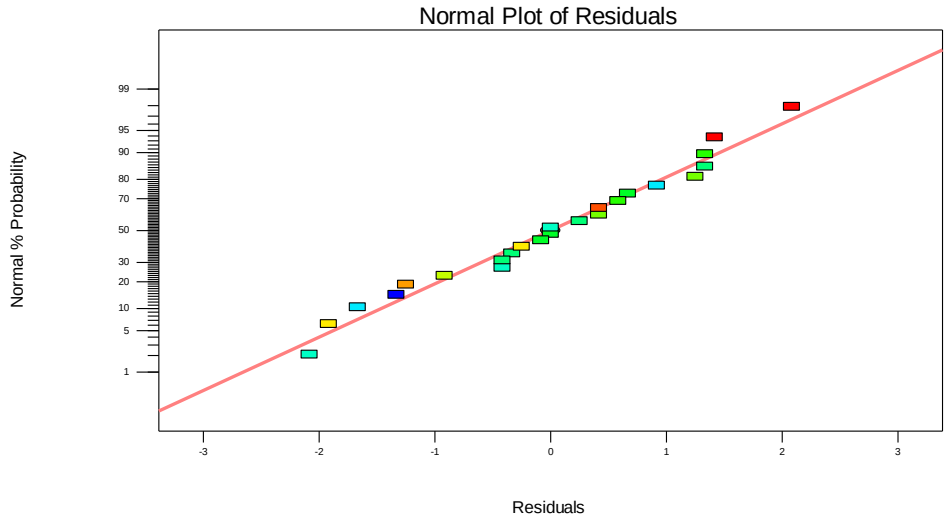
Std. Dev.	1.44	R-Squared	0.8653
Mean	91.83	Adj R-Squared	0.8461
C.V. %	1.57	Pred R-Squared	0.6042
PRESS	85.22	Adeq Precision	11.751

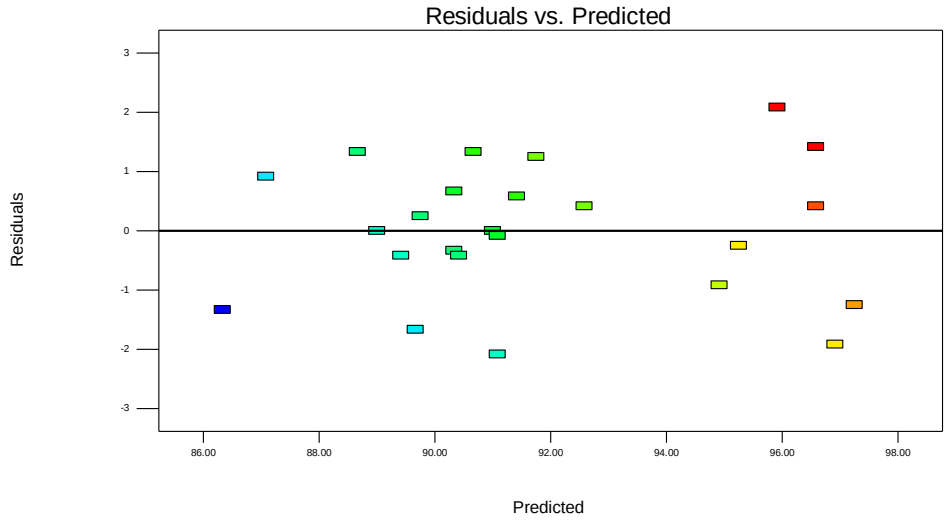
Treatment Means (Adjusted, If Necessary)

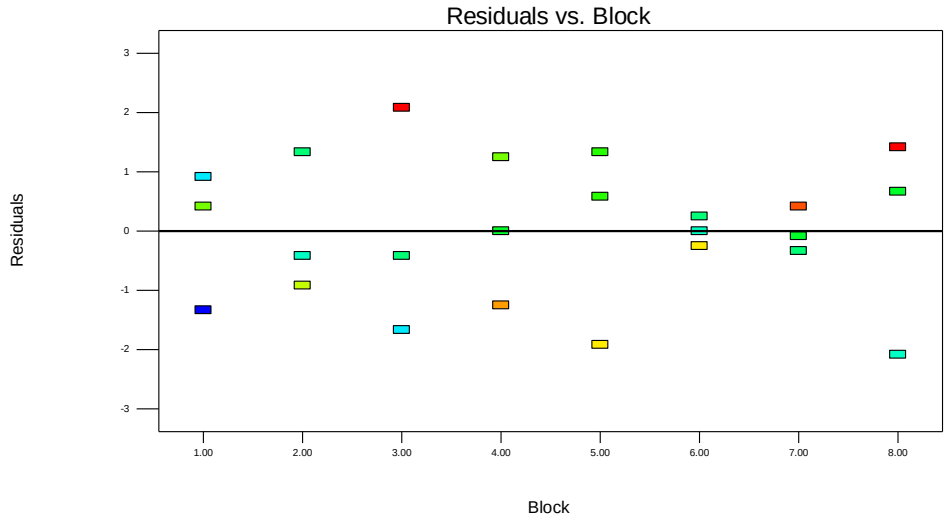
	Estimated Mean	Standard Error
1-Wine 1	89.50	0.51
2-Wine 2	90.25	0.51
3-Wine 3	95.75	0.51

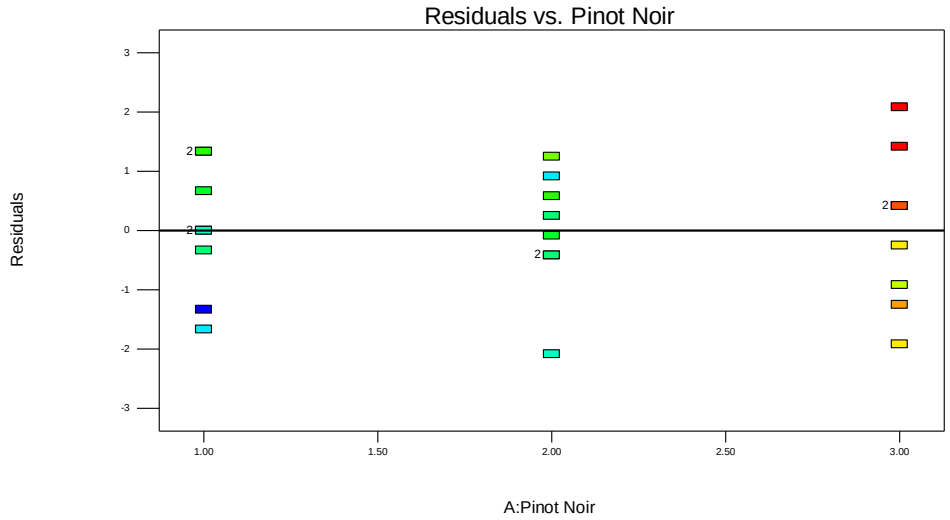
Treatment	Mean Difference	df	Standard Error	t for H0 Coeff=0	Prob > |t|
1 vs 2	-0.75	1	0.72	-1.04	0.3150
1 vs 3	-6.25	1	0.72	-8.69	< 0.0001
2 vs 3	-5.50	1	0.72	-7.64	< 0.0001




Normal Plot of Residuals
Normal % Probability
99
95
90
80
70
50
30
20
10
5
1
-3
-2
-1
0
1
2
3
Residuals


Residuals vs. Predicted
Residuals
3
2
1
0
-1
-2
-3
86.00
88.00
90.00
92.00
94.00
96.00
98.00
Predicted


Residuals vs. Block
Residuals
3
2
1
0
-1
-2
-3
1.00
2.00
3.00
4.00
5.00
6.00
7.00
8.00
Block



15.17. A transformation that stabilizes the variance of the response also may make the response distribution closer to normal.

True False

15.18. A primary reason for inequality of variance is non-normality of the response distribution.

True False

15.19. Inequality of variance in an experiment can be caused by operator fatigue, tool wear, or depletion of some chemical reagent.

True False

15.20. A general linear model had the following components: a response distribution, a linear predictor and a link function.

True False

15.21. A commonly-used link function for binomial data is the logistic link.

True False

15.22. The generalized linear model is actually a non-linear model.

True False

15.28. Use of a generalized linear model is often a good alternative to a data transformation on the response.

True False

15.29. A covariate is a nuisance factor that can be easily controlled by the experimenter.

True **False**

15.30. Analysis of covariance is an extension of the ANOVA intended to incorporate the effects of uncontrolled, but measured variables into the analysis.

True False

15.31. An experiment that includes a covariate can improve the precision of the comparisons between controlled factors.

True False